Quiz 21

C20: $a_n = \tan^{-1}\left(\frac{10n}{10n+4}\right)$

Divergence test:-

$$\lim_{n\to\infty} \tan^{-1}\left(\frac{10n}{10n+4}\right) = \tan^{-1}\left(\frac{\infty}{\infty}\right) = \frac{\pi}{2} \neq 0$$

∴ $\tan^{-1}\left(\frac{10n}{10n+4}\right)$ is Diverges by Divergence test.

(C21)

$$\sum_{n=3}^{\infty} \left(\pi/4\right)^n$$

$$\sum_{n=3}^{\infty} a_n = \sum_{n=3,}^{\infty} \left(\pi/4\right)^n$$

$|r| = \pi/4 < 1$

∴ Converge by Geometric Series

(C21) (A)

$$\sum_{n=1}^{\infty} \frac{5}{3\sqrt{n^4}} = \sum_{n=1}^{\infty} \frac{5}{n^{4/3}}$$

P-Series $\quad p = 4/3 > 1$

∴ $\sum_{n=1}^{\infty} \frac{5}{3\sqrt{n^4}}$ is Converges by P-Series.

(B) $\sum_{n=1}^{\infty} \frac{3n^2+12}{3n-2n^2}$ $\Rightarrow$ Compare to $\sum_{n=1}^{\infty} \frac{3n^2}{2n^2} = \sum \frac{3}{2}$

$\therefore$ Diverges by Divergence test

$\therefore \sum_{n=1}^{\infty} \frac{3n^2+12}{3n-2n^2} > \sum_{n=1}^{\infty} \frac{3n^2}{2n^2}$

$\therefore$ Diverges by comparison.

(C) $\sum_{n=1}^{\infty} \frac{4^{n+1}}{3^n} \Rightarrow \sum_{n=1}^{\infty} 4\left(\frac{4^n}{3^n}\right)$

$= \sum_{n=1}^{\infty} 4\left(\frac{4}{3}\right)^n$

$|r| = \frac{4}{3} > 1$

$\therefore$ Diverges by Geometric Series.

(D) $\sum_{n=1}^{\infty} \left(1 + \frac{8}{n}\right)^n$

Divergence test :-

$\lim_{n \to \infty} \left(1 + \frac{8}{n}\right)^n = 1$

$\therefore$ Diverges by Divergence test

(E) $\sum_{n=1}^{\infty} 1/\sqrt[5]{n^2} = \sum_{n=1}^{\infty} 1/n^{2/5}$

P-Series $p = 2/5 < 1$

$\therefore$ Diverges by p-Series.